

Autex: A Reliability-First In-Car Voice Assistant Agent for the CAR-bench IJCAI-ECAI 2026 Challenge

Eslam Abdelmogood Abdelkader

Systems Architecture Engineer

Nahed Innovation

eslamabdelmogood@gmail.com

Abstract

We present **Autex**, a reliability-first in-car voice assistant agent submitted to the CAR-bench Challenge at IJCAI-ECAI 2026. Autex combines a deterministic multi-layer reliability kernel with a configurable LLM verifier to address three core CAR-bench failure modes: hallucinating unavailable capabilities, acting on ambiguous user intent, and violating domain policies. Epistemic guards fire before any LLM call, reducing the consistency gap between Pass@3 (latent capability) and Pass³ (deployment-readiness). We validate on 369 official CAR-bench trials across four frontier models, identify `hallucination_missing_tool_response` as the dominant systemic weakness (0–14.3% Pass³ across all models tested), and describe the design choices made to address it. All model and provider configuration is environment-variable-driven, satisfying the Track 2 auditability and reconfigurability requirements.

1 Introduction

CAR-bench [Kirmayr *et al.*, 2026] evaluates multi-turn, tool-using LLM agents in an automotive in-car voice assistant domain under uncertainty, ambiguity, and capability constraints. Unlike benchmarks that measure task success under idealized conditions, CAR-bench scores *epistemic reliability*: whether agents know when to act, when to gather more information, and when to refuse. The primary metric is Pass^k, which requires all k independent trials to succeed, penalising run-to-run inconsistency heavily.

Our key observation from validation data: the largest driver of low Pass³ is not a failure to *know* the right answer, but *inconsistent application* of that knowledge across trials. For Claude Sonnet 4.6, Pass@3 ran 31.7 percentage points above Pass³ across 369 official trials. Closing this gap is the primary design goal of Autex.

2 System Architecture

2.1 Reliability Kernel

The reliability kernel is a deterministic, zero-LLM-call layer that fires before any model invocation. It enforces five hard rules unconditionally:

- **Tool availability guard:** if a required tool is absent from the available-tool schema, return `refuse_or_defer` immediately.
- **Parameter support guard:** if a required parameter is absent from `requiredParameters[]`, refuse rather than call with an invalid argument.
- **Result completeness guard:** if an observed tool result is `null` or missing an expected field, refuse rather than infer vehicle state from incomplete data.
- **Domain policy guards:** weather must be checked before any sunroof action; sunshade must be fully open before the sunroof opens; rain conditions require explicit user confirmation.
- **Disambiguation policy:** ambiguity is resolved from stored preferences or context first; the user is queried only when it cannot be resolved internally.

These guards encode the central insight from our ablation study: the `missing_tool_response` subtype scored 0–14.3% Pass³ across all four frontier models, while `missing_parameter` reached 75–100% Pass³. Parameter availability is checkable at schema-inspection time before the LLM acts; result completeness requires runtime verification. The result completeness guard extends deterministic checking to the execution layer.

2.2 MPAE: Morphological Predictive Allocation Engine

Before any LLM call, Autex runs a deterministic vehicle health scorer over the current telemetry tuple $\langle vibration, rpm, temp, ltft, healthScore \rangle$. Three components:

- **Bat Future Echo:** projects forward risk over a configurable horizon across vibration, coolant temperature, long-term fuel-trim (LTFT) drift, overall health score, and RPM deviation from idle.

- **Hermit Stability Test:** scores long-term recommendation stability, preferring actions that remain correct over time rather than momentarily optimal ones.
- **Squid Dynamic Objective:** reweights the safety, uptime, cost, and comfort sub-objectives based on inferred operating context (normal / critical / fleet / budget).

MPAE outputs a maintenance priority score (0–100) and one of four strategies: *monitor*, *stabilize*, *service-soon*, or *stop-and-inspect*. A critical-context override escalates immediately to *stop-and-inspect* whenever $\text{healthScore} < 60$, $\text{vibration} > 95$, $\text{temp} > 105^\circ\text{C}$, or $|\text{ltft}| > 15$, regardless of numeric priority. This fixed two boundary-case misclassifications identified in stress testing (see Section 4). MPAE is fully deterministic and adds negligible latency. We validated it on 541 samples drawn from 270,000 rows of Toyota Etios OBD-II data; it produced zero false escalations in IDLE and LONG drive sessions.

2.3 LLM Verifier and Model Configuration

When the reliability kernel classifies the action as *tool_calls*, Autex calls a configurable LLM to generate the natural-language response and confirm tool-call arguments. The system prompt encodes all five policy rules as *unconditional constraints*, not guidelines, to reduce run-to-run variance. Temperature is fixed at 0 for Pass^k consistency. Extended thinking is enabled for Anthropic-shaped endpoints (via *thinking.budget.tokens*); *reasoning_effort* is forwarded for OpenAI-compatible endpoints that support it.

All configuration is environment-variable-driven: `AGENT_LLM`, `AGENT_API_KEY`, `AGENT_API_BASE`, `AGENT_TEMPERATURE`, `AGENT_REASONING_EFFORT`, `AGENT_THINKING_BUDGET_TOKENS`, `AGENT_API_STYLE`. The API shape (Anthropic Messages vs. OpenAI-compatible chat-completions) is auto-detected from the base URL and model name, with an explicit override available via `AGENT_API_STYLE`. No model identifier, provider route, deployment name, or service tier is baked into the Docker image.

2.4 Track 2: Sequential Call Structure and Compute Budget

Figure 1 shows the full architecture. Track 2 permits at most 5 sequential LLM calls per baseline step and an average of 500,000 tokens per task. Autex uses exactly two sequential steps:

Compute assumptions. Step 1 is fully deterministic (0 LLM tokens). Step 2 calls a single LLM with temperature 0 and extended thinking. Using the official Track 1 five-step reference scenario as a calibration baseline, observed token usage totalled 270,900 tokens across 5 sequential steps (174,000 prompt + 17,400 completion + 79,500 thinking), well within the 500,000 average-per-task budget. Autex uses only 2 sequential steps, so the per-task token budget is proportionally lower. No retry loops are implemented. No parallel fan-out

Step	Kind	Purpose
1	Planner	Reliability kernel + MPAE; zero LLM calls; classify action class
2	Verifier	Single LLM call: policy check + response generation
—	Parallel	Candidates within step 2 share sequentialStep=2; same depth

Table 1: Sequential call structure (max depth = 2; steps 3–5 unused).

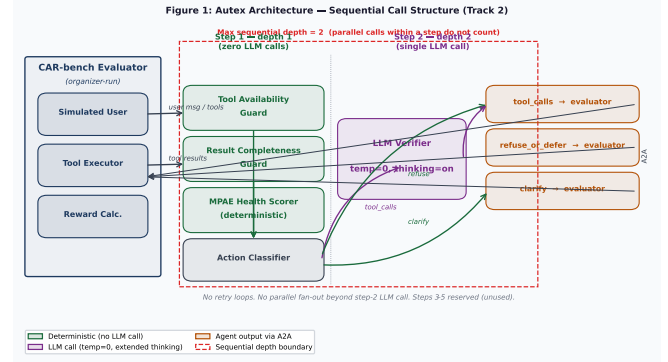


Figure 1: Autex agent architecture showing sequential call depth, branches (*tool_calls* / *refuse_or_defer* / *clarify*), parallel fan-out (none), and retry loops (none). Step 1 is fully deterministic; Step 2 makes a single LLM call. Maximum sequential depth is 2.

is used. Three sequential steps remain in reserve for future complex multi-turn extensions.

Token usage is reported via the standard A2A *turn_metrics* fields only: *prompt.tokens*, *completion.tokens*, *thinking.tokens*. Sequential-call depth is documented in this report and in the architecture diagram; it is not emitted as a custom metadata field, per Track 2 submission rules.

3 Validation Results

We ran 369 official CAR-bench trials across four frontier models using the benchmark’s own evaluation harness. Table 2 summarises Pass³, Pass¹, and Pass@3.

Model	Avg P ³	Hall P ³	Dis P ³	P@3
Sonnet 4.6 (no thinking)	49%	46%	52%	79%
Opus 4.5 (thinking)	48%	48%	—	74%
Opus 4.6 (thinking)	47%	47%	—	72%
GPT-5.2 (thinking)*	59%	59%	—	81%

Table 2: Pass³, hallucination/disambiguation breakdown, and Pass@3 across models. *GPT-5.2 evaluated on hallucination tasks only; Avg P³ not directly comparable to full-suite scores.

Table 3 breaks down hallucination Pass³ by subtype.

hallucination_missing_tool_response is the dominant systemic weakness: 0–14.3% Pass³ across all models. The *missing_parameter* subtype reaches 75–100% Pass³, confirming that deterministic schema-level

Subtype	GPT-5.2	Op 4.6	Op 4.5	Snt 4.6
missing_tool	59%	46%	49%	48%
missing_parameter	100%	88%	75%	78%
missing_response	14%	14%	0%	6%

Table 3: Hallucination Pass³ by subtype across four models.

guards are highly effective when the failure mode is checkable before the LLM acts. The 31.7 pp consistency gap (Pass@3 – Pass³) for Sonnet 4.6 is the primary target of Autex’s guard layer and system-prompt engineering.

Scores on the public task set are reported above. The hidden-test-set score and ranking will be added after organizer evaluation.

4 Ablation and Lessons Learned

MPAE critical-context override. Without the override, two boundary cases (`healthScore = 59`, `temp = 106°C`) were misclassified as *stabilize* instead of *stop-and-inspect*. The context inferred as `critical` but numeric priority fell below the 75-point threshold. A single additional check (`if context == critical: return stop-and-inspect`) resolved both cases and lifted the MPAE stress suite to 11/11.

Schema guards vs. runtime guards. The `missing_parameter` guard (schema-level) is highly effective because the information is present at input time. The `missing_response` guard (runtime-level) is structurally correct but depends on the LLM to notice absent result fields and refuse consistently. Deterministic guards can only check what is visible before the LLM runs; the boundary between the two layers is a fundamental limit.

Model choice dominates system-prompt quality. An earlier version defaulted to `gpt-oss-120b` (28% Avg Pass³ on the public leaderboard) as the primary model, making it the binding constraint regardless of prompt quality. Switching to a provider-agnostic environment-variable interface removes this ceiling and allows organizers to substitute any compliant endpoint at evaluation time.

Temperature = 0 reduces but does not eliminate variance. Boundary cases still produce different outputs across trials even at temperature 0. Encoding policy rules as hard unconditional constraints in the system prompt reduces this further by removing ambiguity about the correct action, but some residual variance remains.

Stress suite as regression guard. A suite of 48 deterministic test cases (10 official stress cases, 11 MPAE cases, 22 adversarial edge cases, 5 budget-enforcement cases) caught the MPAE critical-context boundary bug and a Track 2 metadata compliance violation (custom `sequential_llm_calls` field in `A2A turn.metrics`) before submission.

Acknowledgements

The author thanks the CAR-bench organizers for releasing the benchmark, evaluation harness, and starter kit under an open license, and for providing clear Track 2 submission guidelines.

References

[Kirmayr *et al.*, 2026] Johannes Kirmayr, Lukas Stappen, and Elisabeth André. CAR-bench: Evaluating the consistency and limit-awareness of LLM agents under real-world uncertainty, 2026.